

Pabna University of Science and Technology



Department of Statistics

Course Title: Statistical Field Survey

Course Code: STAT-4110

Statistical Field Survey

On

“Impact of Heatwaves on Human Health: A Case Study in Pabna”

Submitted by

Roll: 181632

Session: 2017-18

4th Year 1st Semester

Dept. of Statistics

Pabna University of Science and Technology

Date of Submission: 20-11-2023

Supervised by:

Md. Shahajada Mia

Assistant Professor

Dept. of Statistics

PUST

DECLARATION

I declare that this survey is an original work submitted by the following group members who have all activity made a contribution. This survey has not been submitted for any degree or examination at any other university. It does not contain any other person's data, pictures, graphs or other information, unless specifically acknowledged as being sourced from those persons. Any other work of similar nature has been appropriately referenced in this survey.

Signed:

Roll: 181632 Reg. No: 1165106

CERTIFICATE OF APPROVAL

This is to certify that the survey titled: "**Impact of Heatwaves on Human Health: A Case Study in Pabna**"

Conducted by-

Roll: 181632

has been reviewed and approved as satisfactory in partial fulfillment of the requirement of the academic research of the degree of B.Sc in Statistics.

Md. Shahajada Mia

Assistant Professor

Dept. of Statistics

Pabna University of Science and Technology

ACKNOWLEDGEMENT

All praise and glory to Almighty Allah who has given us the strength, patience and ability to make this work possible.

I express sincere gratitude to our supervisor **Md. Shahajada Mia**, Assistant Professor, Dept. of Statistics, Pabna University of Science and Technology for his incredible guidance and support throughout the process of conducting this survey.

I also would like to express our love to our respected teachers **Dr. Md. Shamim Reza, Dr. Sabba Ruhi** and **Most. Mira Khatun**, and the entire Department of Statistics, Pabna University of Science and Technology for giving us valuable suggestions and ideas.

All in all, I would like to thank everyone involved in this survey and helped us with their suggestions to make the survey better.

Finally, I would like to thank our parents and friends for always being with us and supporting us in every situation.

November 2023

Table of Contents

Preliminary Part

Contents	Page
Title	i
Declaration	ii
Certificate of approval	iii
Acknowledgement	iv
Abstract	ix

Chapter One

Contents	Page
Introduction	1
Objectives	1
Literature Review	2
Research Gap	3

Chapter Two

Contents	Page
Data Collection method	4
Time of data collection Study area of data collection	4
Study area of data collection	4
Sampling Method Sample Size Determination	4
Sample Size Determination	5
Data Collection	5
Data Analysis	6
Ethical Considerations	6

Chapter Three

Contents	Page
Data Analysis and Presentation	7
Frequency distribution and Graphical Representation	7-32
Associations between various Quantitative Variables	33
Associations between various Qualitative Variables	34
Model Fitting	37

Chapter Four:

Contents	Page
Summary and Conclusion	39
Further Study	40
Application	40
Limitation	41
Reference	42
Appendix	43

List of Table

Table No. and Name	Page No
Table-3.1: Frequency distribution of male and female respondents	8
Table-3.2: Frequency distribution of level of education of respondents	9
Table-3.3: Frequency distribution of marital status of respondents	10
Table3.4: Frequency distribution of Socio-economic status of respondents	11-12
Table 3.5: Frequency distribution of BMI of the respondents	13
Table-3.6: Frequency distribution of Health changes status of respondents	14
Table-3.7: Frequency distribution of physical health issues during heatwaves of respondents	15
Table-3.8: Frequency distribution of categories of Physical health issues during heatwaves	16-17
Table-3.9: Frequency distribution of mental health issues during heatwaves of respondents	18-19
Table-3.10: Frequency distribution on impact of sleeping patterns during heatwaves	20
Table-3.11: Frequency distribution of eating habits during heatwaves	21-22
Table-3.12: Frequency distribution of water consumption during heatwaves	23
Table-3.13: Frequency distribution of prioritize dehydration and consume more fluids	24
Table-3.14: Frequency distribution in performance of work productivity	25
Table-3.15: Frequency distribution of support or accommodations from their employer to cope of heatwaves	26
Table-3.16: Frequency distribution of effect of heatwaves on monthly income	27
Table-3.17: Frequency distribution of Financial challenges during heatwaves	28
Table-3.18: Frequency distribution of Financially prepared to handle unexpected expenses related to heatwaves	29
Table-3.19: Frequency distribution of Colling facilities of respondent	30

Table-3.20: Frequency distribution of Categories of Colling facilities	31-32
Table-3.21: Pearson Correlation table	33
Table-3.22: Table for Chi-Square Tests	34
Table-3.23: Table for Chi-Square Tests	35
Table-3.24: Table for Chi-Square Tests	36

List of Figure

Figure No. and Name	Page No
Figure 3.0 Figure 3.0 age of respondents	7
Figure 3.1 Proportion of male and female	8
Figure 3.2 Proportion of level of education	9
Figure 3.3 Marital status of respondents	10
Figure 3.4 Proportion of Socio economic status	11
Figure 3.5 Working time(In hours) of these respondent	12
Figure 3.6 Working time(In hours) of respondents	13
Figure 3.7 Health changes for heatwaves than normal weather of respondents	14
Figure 3.8 Physical health issues of these respondents	15-16
Figure 3.9 Categories of Physical health issues during heatwaves	17
Figure 3.10 Mental health issues of respondents	18
Figure 3.11 Mental health issues of respondents	19
Figure 3.12 Impact on sleeping patterns	20
Figure 3.13 changes in eating habits	21
Figure 3.14 Types of food consume more frequently	22
Figure 3.15 water consumption during heatwaves	23
Figure 3.16 Prioritize Dehydration and consume more fluids	24
Figure 3.17 Performance in work productivity	25
Figure 3.18 support or accommodations from their employer to cope of heatwaves	26
Figure 3.19 Effect of heatwaves on monthly income	27
Figure 3.20 Effect of heatwaves on monthly income	28
Figure 3.21 Effect of heatwaves on monthly income	29
Figure 3.22 Effect of heatwaves on monthly income	30
Figure 3.22 Cooling facilities	31

Abstract:

This survey investigates the multifaceted impact of heatwaves on human health in Pabna, exploring a diverse range of demographic, health, lifestyle, and socio-economic factors. A sample of 300 respondents provided insights into the challenges faced during heatwaves on both physical and mental health issues. The findings underscore the prevalence of dehydration and its intersections with other health concerns. Mental health, notably anxiety and decreased focus, emerges as a significant dimension affected by heatwaves.

Demographically, the sample is predominantly male (75.34%), with diverse educational backgrounds, occupations, and socio-economic statuses. The statistical breakdown showcases that the majority of respondents fall into the middle socio-economic category (45%).

Healthwise, 96.67% of respondents reported health changes during heatwaves, emphasizing the pervasive impact of climatic conditions on well-being. Physical health issues, dominated by dehydration (23.67%), fever (6%), and skin problems (5%), are statistically dissected to reveal insightful patterns.

Mental health emerges as a critical dimension, with 92% reporting issues during heatwaves. Anxiety (57.9%) and decreased focus (54.5%) are predominant, substantiating the intersection of climatic factors and mental well-being.

The impact extends to lifestyle, as 67% of respondents experienced changes in sleeping patterns. Alterations in eating habits are statistically significant, with 50% eating less during heatwaves.

Financial repercussions are scrutinized, revealing that 38% of respondents experience a decrease in monthly income. Furthermore, 87.67% face increased utility bills during heatwaves.

Employer support during heatwaves is statistically distributed, with 58.33% receiving frequent breaks and access to shaded areas. The nuanced relationship between employer support and occupation is teased out statistically.

In conclusion, this survey contributes a holistic understanding of the impact of heatwaves on human health, encompassing physical, mental, and socio-economic dimensions. The findings provide a valuable foundation for targeted interventions and policy considerations to mitigate the adverse effects of heatwaves on the health and well-being of the population in Pabna.

CHAPTER ONE

Introduction:

A prolonged period of abnormal hot weather, known as heatwaves, has become an increasingly prevalent and concerning global climate phenomenon. Heatwaves have severe implications for human health and well-being, posing significant challenges to vulnerable populations, particularly in urban centers. As the earth's temperatures rise due to climate change, the frequency, intensity, and duration of heatwaves are expected to increase, making them a critical public health concern.

One such vulnerable urban center is Pabna City, located in the northwest region of Bangladesh. This densely populated city experiences intense heatwaves during certain periods of the year, impacting the health and daily lives of its residents. As the effects of climate change continue to unfold, understanding the specific risks and impacts of heatwaves on human health in Pabna City is of absolute importance.

The impact of heatwaves on human health is a pressing research area that demands urgent attention to develop effective strategies to safeguard public health during extreme heat events. Previous studies have shown that heatwaves can lead to a range of health issues, from mild heat-related illnesses like heat rashes to severe conditions such as heat exhaustion and heatstroke. Additionally, heatwaves have been associated with exacerbating existing health problems, including respiratory and cardiovascular diseases, kidney disorders, and mental health challenges.

While the health impacts of heatwaves are evident, the vulnerability and coping strategies of individuals in Pabna City require further exploration. The unique geographical context and local socio-economic factors may influence the health effects experienced by the city's diverse population. Understanding these dynamics will enable the development of targeted interventions and policies to protect vulnerable groups effectively.

This study aims to investigate the "Impact of Heatwaves on Human Health: A Study in Pabna City, Bangladesh." By delving into the health effects of heatwaves in this specific urban context, the study endeavors to contribute to the existing body of knowledge on climate change-related health risks, particularly concerning extreme temperatures.

Objectives:

- To find out the impacts of heatwaves on human physical health and mental health.
- To determine the change of food habit during heatwaves.
- To find out the change of work performance and work productivity during heatwave.
- To assess the financial stability of outdoor Workers during heatwaves.

Literature Review:

It is widely accepted that increased exposure to heat has a detrimental effect on human health, resulting in increased mortality (death) and morbidity (illness) across a variety of geographical locations (Anderson and Bell, 2011, Haines et al., 2006, Loughnan et al., 2010, Martiello and Giacchi, 2010, Zeng et al., 2016)).

This effect has been demonstrated by a number of extreme heat events worldwide, including in Chicago, USA in 1995 (Whitman et al., 1997), the European-wide heatwave in 2003 (Kosatsky, 2005) and in South-East Australia in 2009 (Nitschke et al., 2011). Furthermore, a relationship between increasing temperature and increasing mortality and morbidity has been found across several global locations (Michelozzi et al., 2009, Stafoggia et al., 2006, Sugg et al., 2016, Sung et al., 2013), demonstrating that heat illness and death can occur irrespective of absolute temperature and outside a designated extreme event.

While all sectors of the population are at risk of illness and death when exposed to increased heat, and especially extreme heat, particular sub-groups are more vulnerable than others. However, relationships between temperature and health impacts are neither uniform nor predictable, influenced by a number of complex and interacting factors including biological, environmental, medical, social and geographical factors (Klinenberg, 2002, Uejio et al., 2011, Yihan et al., 2013).

The impacts of heatwaves on human health are influenced by the vulnerability and exposure of the affected population. Vulnerable populations may experience more significant health risks during heatwaves (HEAT WAVES AND HUMAN HEALTH, February 2019). Heatwaves can lead to increased mortality rates, and this is evident from past events, such as the European heatwave of 2003 and the Central European and Russian heatwave of 2010. Extreme heat events can have disastrous consequences for human populations, especially in regions where the population is not adequately prepared to cope with high temperatures (The impact of heat waves on mortality in Northwest India .Environmental Research, Volume 176, September 2019, 108546). Rising global ambient temperatures affect all populations. However, some populations are more exposed to, or more physiologically or socio-economically vulnerable to physiological stress, exacerbated illness, and an increased risk of death from exposure to excess heat. These include the elderly, infants and children, pregnant women, outdoor and manual workers, athletes, and the poor. Gender can play an important role in determining heat exposure. WHO (World Health Organization) 1 June 2018

Based on these sources, it is evident that heatwaves can have significant and wide-ranging impacts on human health, with vulnerable populations being more susceptible to adverse effects.

Research Gap:

- **Geographical Context and Local Vulnerability:** Most of the existing research on the health impacts of heatwaves has been conducted in developed countries or larger urban areas. There is a lack of studies that focus on the unique characteristics of smaller cities or urban centers in developing countries like Pabna City, Bangladesh. The research gap lies in the need to understand how the geographical context and local vulnerability factors influence the health effects of heatwaves in this specific city.
- **Inadequate understanding of how socio-economic factors contribute to health disparities during heatwaves in Pabna City.**
- **Lack of knowledge on the effectiveness and feasibility of heatwave adaptation strategies and community resilience in this specific context.**
- **Insufficient exploration of heatwave awareness and behavior change among residents of Pabna City.**
- **Scarcity of studies examining the long-term health impacts of recurrent heatwave exposure in the region.**

CHAPTER TWO

Methodology

Data Collection method:

Data collection is the process where relevant data are collected from the respondents or any sources by applying various method. It's considered as one of the most important task in the research process. In modern age, research is going to be based on secondary data but reliability and accuracy of any result primary data can't overlooked. There are two types of data such as qualitative data and quantitative data. In our survey study we used primary data collection method. The primary data collection method are classified into two parts based on the nature of the data such as "Quantitative data and Qualitative data".

Time of data collection:

We have collected whole required data during 27 September,2023 to 27 October, 2023.

Study Area of Data Collection:

The study will be conducted in Pabna City, Bangladesh, due to its vulnerability to heatwaves and the relevance of the research topic to its urban population.

Sampling Methods:

There are several different sampling techniques available, and they can be divided into two groups: (i) probability sampling, (ii) non-probability sampling.

Probability Sampling Methods:

Probability Sampling is a sampling technique in which sample from a larger population are chosen using a method based on the theory of probability. For a participant to be considered as a probability sample, he/she must be selected using a random selection. Mainly there are four types of probability sampling:

1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Clustered sampling

Non-Probability Sampling Methods

Non-probability sampling is a sampling technique in which the researcher selects samples based on the subjective judgment of the researcher rather than random selection. Mainly there are four types of non- probability sampling.

In our survey, we conduct the probability sampling, simple random sampling.

Sample Size Determination:

The Cochran formula is commonly used in statistical research to calculate the sample size needed for estimating the true proportion of a population with a specified level of precision. This formula is particularly relevant when dealing with large populations.

The Cochran formula for sample size determination is given by:

$$n = \frac{Z^2 * p * q}{e^2}$$

Where:

n is the required sample size.

Z is the Z-score corresponding to the desired level of confidence (e.g., 1.96 for a 95% confidence level).

P is the estimated proportion of the population.

e is the margin of error (desired level of precision).

Let

$p=0.6$ (estimated proportion)

$Z=1.96$ (Z-score for a 95% confidence level)

E (margin of error), which we calculated as approximately 0.0554

After calculating Cochran formula we get

$$n \approx 299.999$$

A sample size of 300 individuals between the ages of 15 to 80 years. Samples are selected through simple random sampling to ensure representation across various age groups.

The participants are included outdoor workers, businessmen, construction workers, rickshaw pullers, and others from different socio-economic backgrounds.

Data Collection:

Quantitative Data Collection:

A structured questionnaire is used to collect quantitative data, covering demographics, physical health issues experienced during heatwaves, mental health impacts, changes in eating and sleeping patterns, water consumption, work performance, and financial challenges.

Data on participants' monthly income and access to cooling facilities is also collected.

Qualitative Data Collection:

In-depth interviews is conducted with a subset of participants to gain deeper insights into their experiences and perceptions regarding heatwave impacts on health, coping strategies, and access to support or accommodations.

Data Analysis:

Quantitative Data Analysis:

Statistical software is used to analyze the quantitative data, employing descriptive statistics to summarize demographic characteristics and health impacts.

Relationships between variables is examined using correlation analysis.

Qualitative Data Analysis:

Thematic analysis is used to identify patterns and themes in the qualitative data obtained from interviews.

The analysis will involve coding and categorizing participants' responses to explore common experiences and perspectives.

Ethical Considerations:

Informed consent is obtained from all participants before data collection, ensuring their voluntary participation.

Participants' privacy and confidentiality will be strictly maintained throughout the survey process.

Chapter Three

Data Analysis and Presentation

3.1 Data Analysis

Data analysis involves actions and methods performed on data that help describe facts, detect patterns, develop explanations and test hypotheses. This includes data quality assurance, statistical data analysis, modeling, and interpretation of results. We use SPSS (Statistical Package for Social Science) and Excel 2016 software for our data analysis.

3.2 Frequency distribution and Graphical Representation

Frequency distributions are visual displays that organize and present frequency counts so that the information can be interpreted more easily. A frequency distribution of data can be shown in a table or graph. Some common methods of showing frequency distributions include frequency tables, histograms or bar charts. A frequency distribution also shows us a summarized grouping of data divided into mutually exclusive classes and the number of occurrences in a class. It is a way of showing unorganized data e.g. to show results of an election, income of people for a certain region, sales of a product within a certain period, student loan amounts of graduates, etc. Some of the graphs that can be used with frequency distributions are histograms, line charts, bar charts and pie charts. Frequency distributions are used for both qualitative and quantitative data.

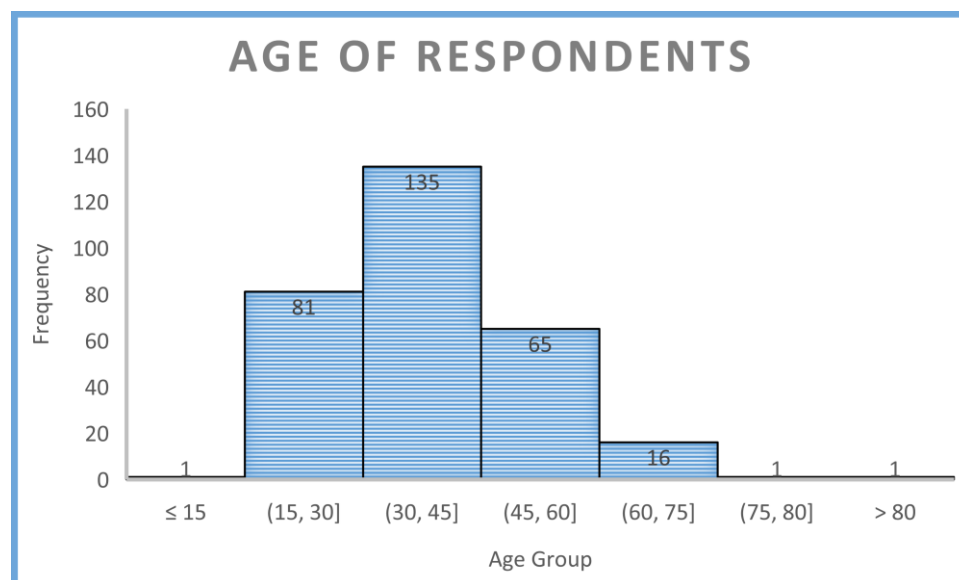


Figure 3.0 age of respondents

Comment: The figure suggests that a significant majority of the surveyed individuals are in the age range of 30 to 45, indicating that this survey are more representative of this age group of 30 to 50.

3.2.1 Proportion of male and female

Table-3.1: Frequency distribution of male and female respondents

Proportion of male and female			
	Frequency	Percent	Cumulative percent
Male	226	75.34	75.33
Female	74	24.67	100
Total	300	100	

3.2.1 The graphical distribution of male and female respondents are given below:

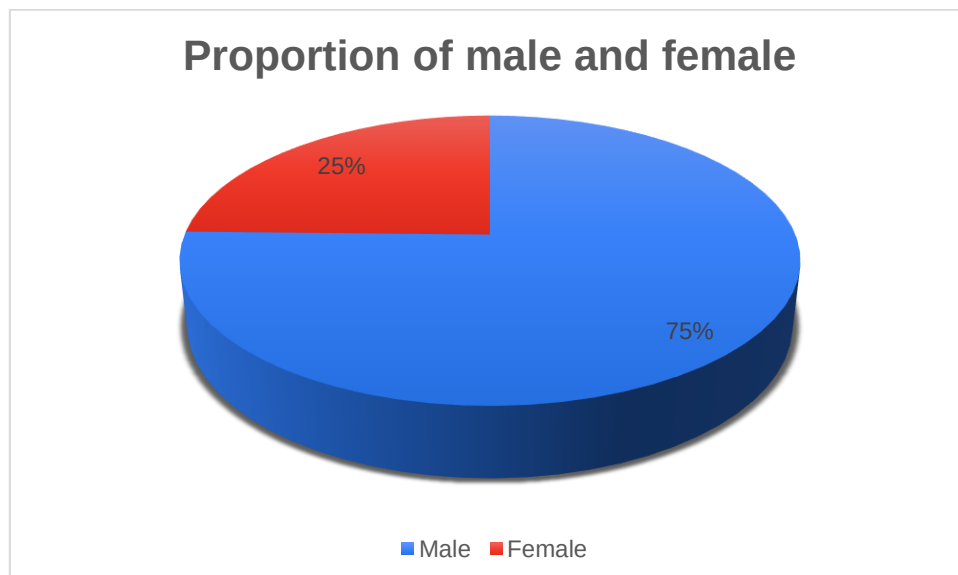


Figure 3.1 Proportion of male and female

Comment: From the above figure we found that total respondents are 300. Among them 75% respondents are male and 25% respondents are female.

3.2.2 Level of education of respondents

Table-3.2: Frequency distribution of level of education of respondents

Level of education	Frequency	Percent	Cumulative Percent
Illiterate	31	10.3	10.3
Primary	93	31.0	41.3
Secondary	63	21.0	62.3
Higher Secondary	37	12.3	74.7
Graduate	58	19.3	94.0
Under Graduate	16	5.3	99.3
Degree	2	0.7	100.0
Total	300	100.0	

The graphical distribution of level of education of respondents are given below:

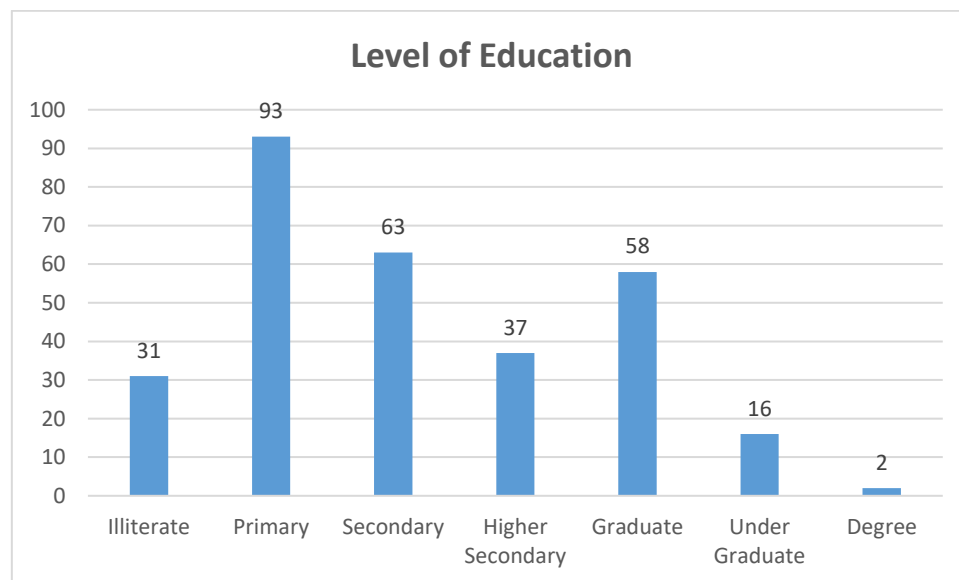


Figure 3.2 Proportion of level of education

Comment: From the above table we found that total Respondents 300. Total of 300 respondents ,most of them are from primary level of education.also we have good amount of secondary and graduate level of education respondents.

3.2.3 Marital Status of respondents

Table-3.3: Frequency distribution of marital status of respondents

Marital Status			
	Frequency	Percent	Cumulative Percent
Married	251	83.67	83.67
Unmarried	46	15	98.67
Divorced	1	0.33	99
Widowed	2	1	100
Total	300	100.00	

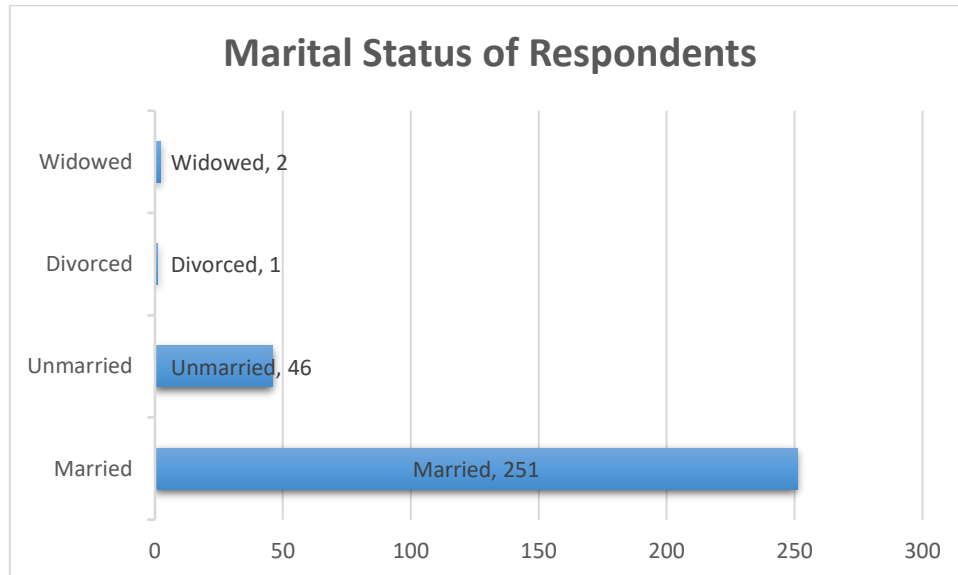


Figure3.3 Marital status of respondents

Comment: From the above table and figure we found that total patients is 300. Among them 83.6 % are married and 15 % are unmarried and so on.

3.2.4 Socio economic status:

Table3.4: Frequency distribution of Socio-economic status of respondents

Socio Economic Status			
	Frequency	Percent	Cumulative Percent
Poorest	30	10	10
Poor	110	36.67	46.7
Middle	135	45	91.7
Rich	24	8	99.7
Richest	1	0.34	100
Total	300	100	

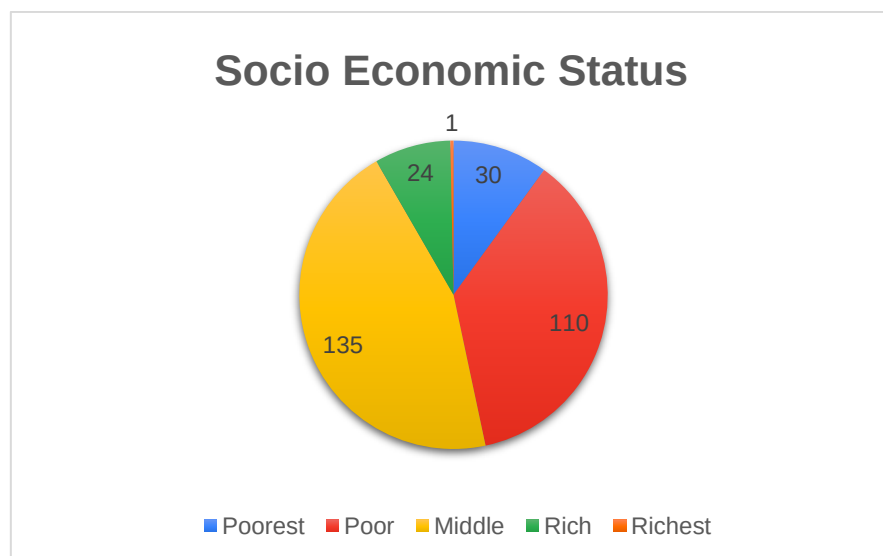


Figure 3.4 Proportion of Socio economic status

Comment: From the above table and figure we found that a significant proportion falls within the middle-income category, comprising 45% of the participants. The poor and poorest categories represent 36.67% and 10%, respectively. Interestingly, a minor percentage falls into the rich and richest categories, indicating a varied socio-economic landscape among our surveyed individuals.

3.2.5 Working time (In hours) of respondents:

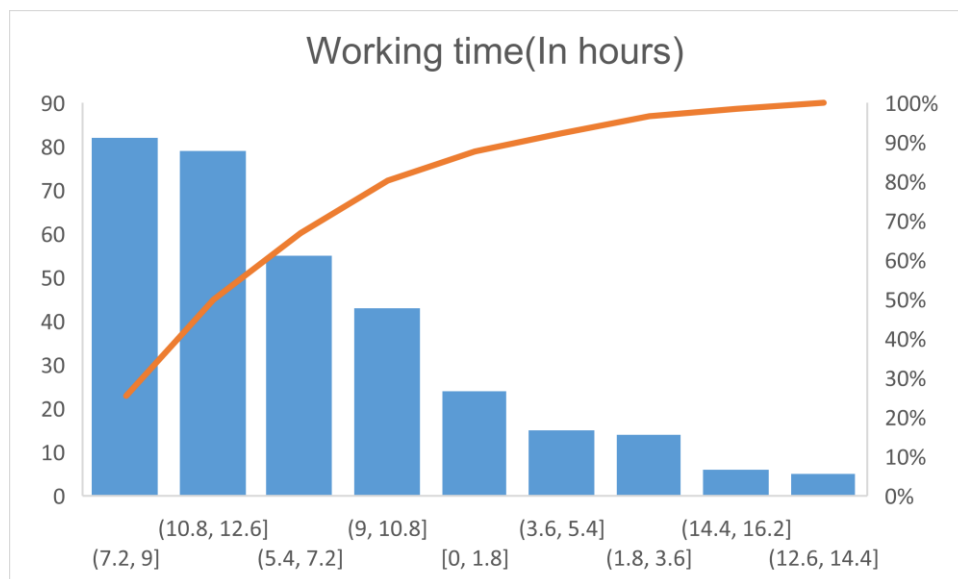


Figure 3.5 Working time (In hours) of these respondent

Comment: From the above figure we found that most of the respondents are work within 7.2-9 hours interval and 10.8-12.6 hours interval.

3.2.6 BMI of Respondent:

Table 3.5: Frequency distribution of BMI of the respondents

BMI of Respondent			
	Frequency	Percent	Cumulative Percent
Obesity	24	8	8
Overweight	64	21.33	29.3
Normal	191	63.66	93
Underweight	21	7	100
Total	300	100	

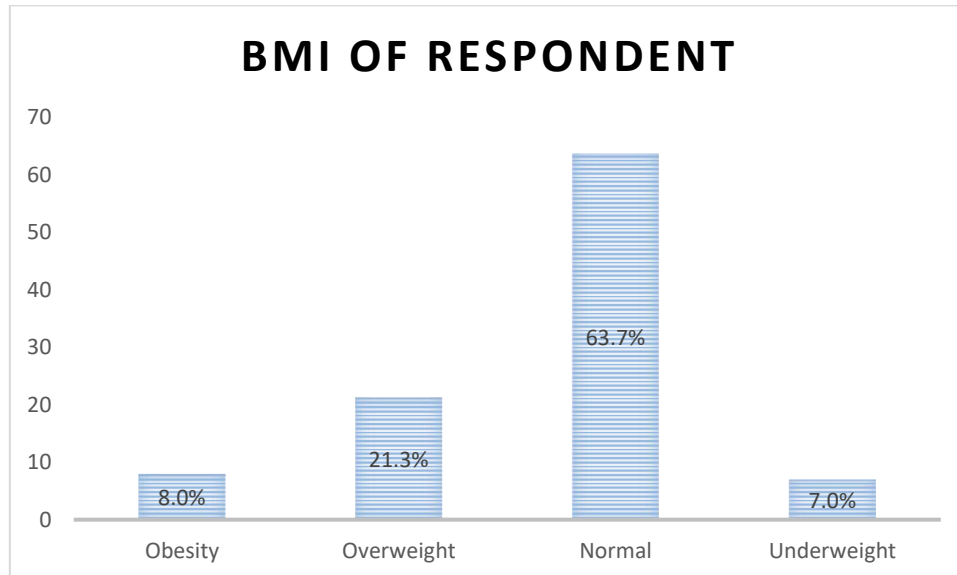


Figure 3.6 Working time (In hours) of respondents

Comment: From the above figure we see that, the majority fall within the 'Normal' range, constituting 63.67% of participants. Approximately 21.33% are categorized as 'Overweight,' while 8% fall under 'Obesity.' Notably, 7% are classified as 'Underweight.' This emphasizes the importance of understanding and addressing varied health needs among the surveyed population."

Specific Analysis

3.2.7 Health changes for heatwaves than from normal weather of respondents

Table-3.6: Frequency distribution of Health changes status of respondents

Health changes for heatwaves than normal weather			
Categories	Frequency	Percent	Cumulative percent
Yes	290	96.67	96.7
No	10	3.34	100
Total	300	100	

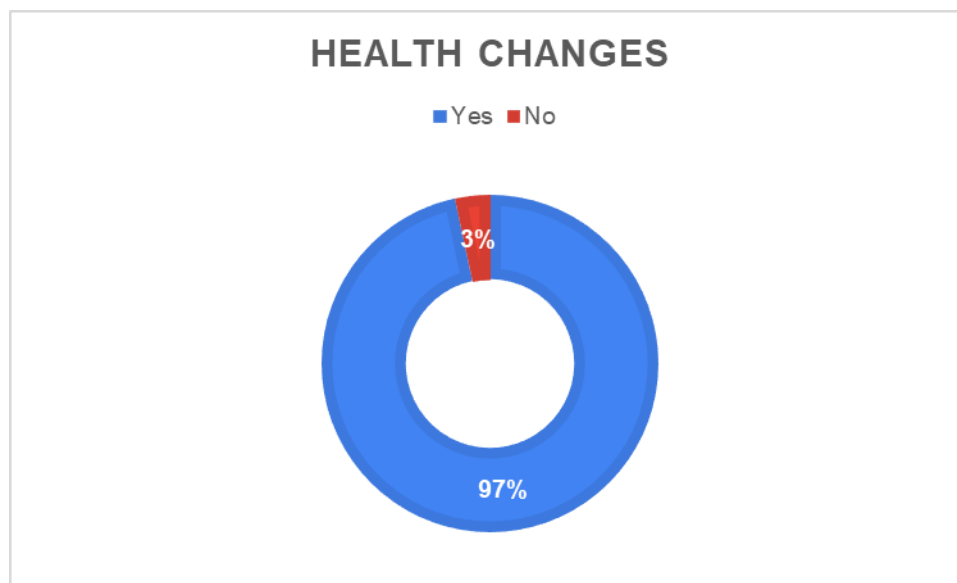


Figure 3.7 Health changes for heatwaves than normal weather of respondents

Comment: The impact of heatwaves on individuals is significant, with 96.67% of the surveyed population experiencing health changes during these periods. This emphasizes the need for a comprehensive understanding of health dynamics in the context of rising temperatures. Only a small percentage (3.34%) reported no discernible changes, suggesting a widespread influence of heatwaves on health conditions among our respondents

3.2.8 Physical health issues during heatwaves:

Table-3.7: Frequency distribution of physical health issues during heatwaves of respondents

physical health issues during heatwaves			
	Frequency	Percent	Cumulative percent
Yes	288	96	96
No	12	4	4
Total	300	100	100

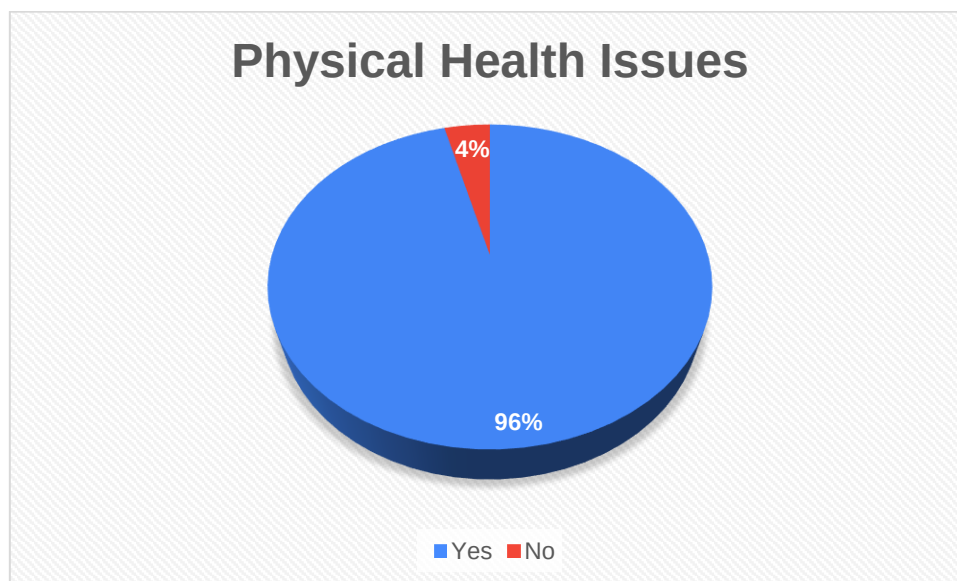


Figure 3.8 Physical health issues of these respondents

Comment:

During heatwaves, 96% of participants faced physical health issues, underscoring the prevalence of challenges. Only 4% reported no such issues, emphasizing the widespread impact on physical well-being.

3.2.9 Categories of Physical health issues during heatwaves:

Table-3.8: Frequency distribution of categories of Physical health issues during heatwaves

Categories	Frequency	Percentage	Cumulative Percentage
Dehydration	71	23.67	23.67
Fever	18	6	29.67
Skin problems	15	5	34.67
Dehydration, Fever	35	11.67	46.3
Dehydration, Skin problems	39	13	59.33
Dehydration, headache	26	8.67	68
Dehydration, Loose bowels	11	3.67	71.67
Dehydration, hypertension	8	2.67	74.33
Dehydration, Heatstroke	5	1.67	76
hypertension, Fever	5	1.67	77.67
Dehydration, Fever, Skin problems	20	6.67	84.33
Dehydration, Heatstroke, Skin problems	3	1	85.33
Dehydration, Fever, Loose bowels	16	5.33	90.67
Dehydration, Skin problems, hypertension	7	2.33	93
Dehydration, Vomit, Skin problems, headache	3	1	94
Dehydration, Fever, Vomit, Skin problems	4	1.33	95.33
Dehydration, Fever, Skin problems, Muscle cramps	2	0.67	96
No change	12	4	100
Total	300	100	

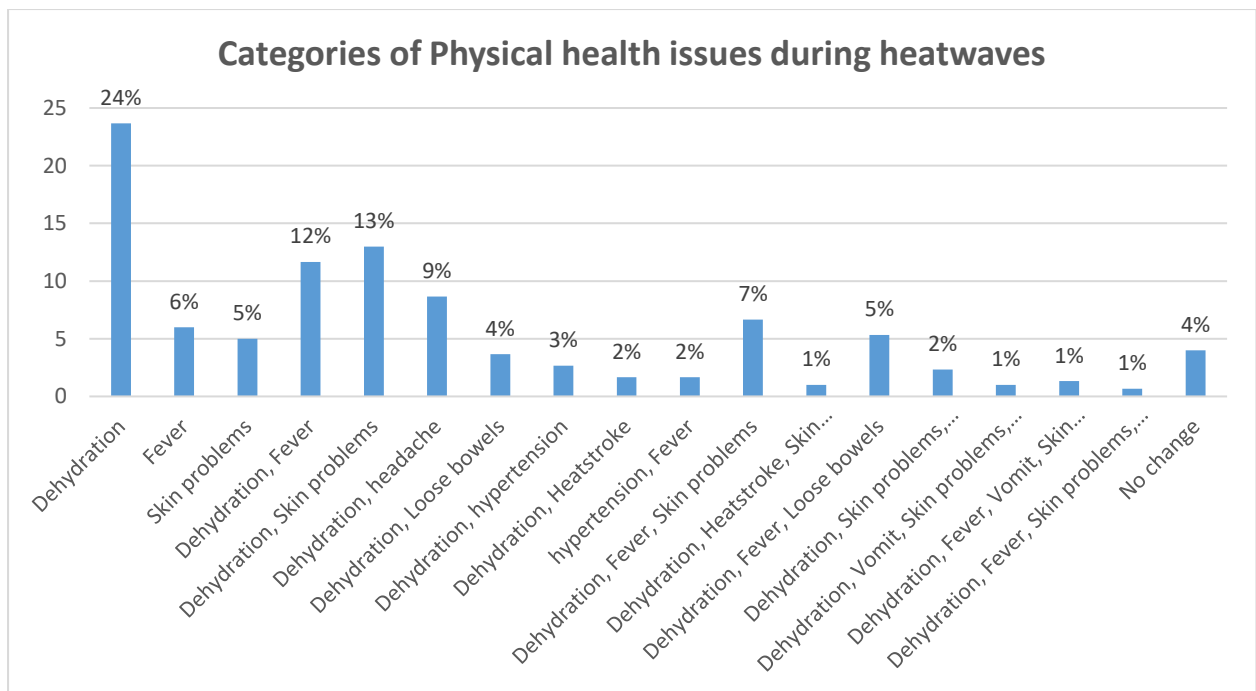


Figure 3.9 Categories of Physical health issues during heatwaves

Comment: The breakdown of physical health issues during heatwaves reveals a diverse range of challenges. Dehydration is the most common, affecting 23.67% of respondents. Various combinations of symptoms, including fever, skin problems, headache, and more, illustrate the multifaceted nature of health impacts during heatwaves. Importantly, 4% reported no physical health changes

3.2.10 Mental Health issues during heatwaves:

Table-3.9: Frequency distribution of mental health issues during heatwaves of respondents

Mental Health issues during heatwaves			
	Frequency	Percent	Cumulative percent
Yes	276	92	92
No	24	8	100
Total	300	100	

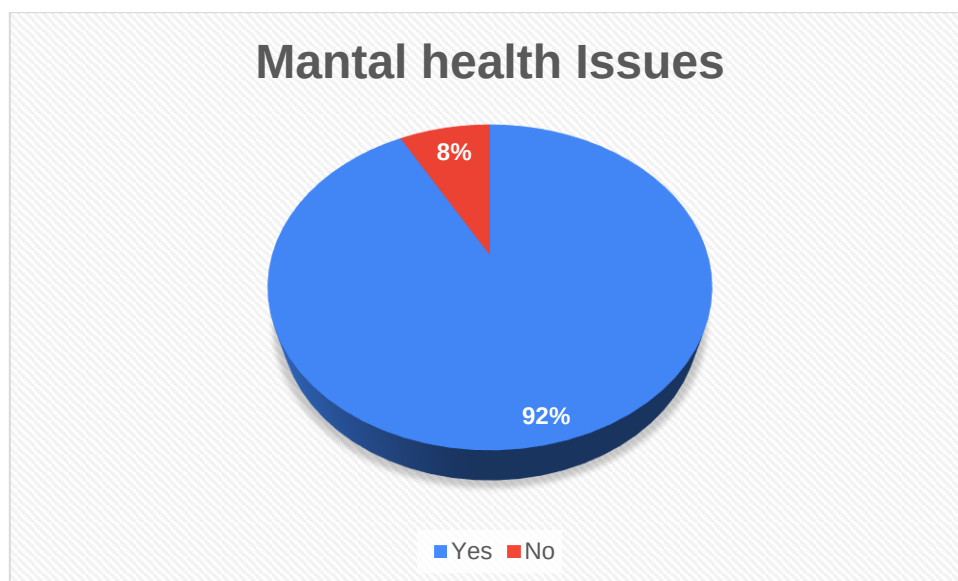


Figure 3.10 Mental health issues of respondents

Comment: The data on mental health issues during heatwaves underscores a significant impact, with 92% of respondents reporting such challenges. This highlights the pervasive nature of mental health concerns in the context of extreme weather conditions. The remaining 8% did not experience noticeable changes in mental well-being during heatwaves.

3.2.11 Types of mental Health issues during heatwaves:

Mental health issues you have experienced during heatwaves:

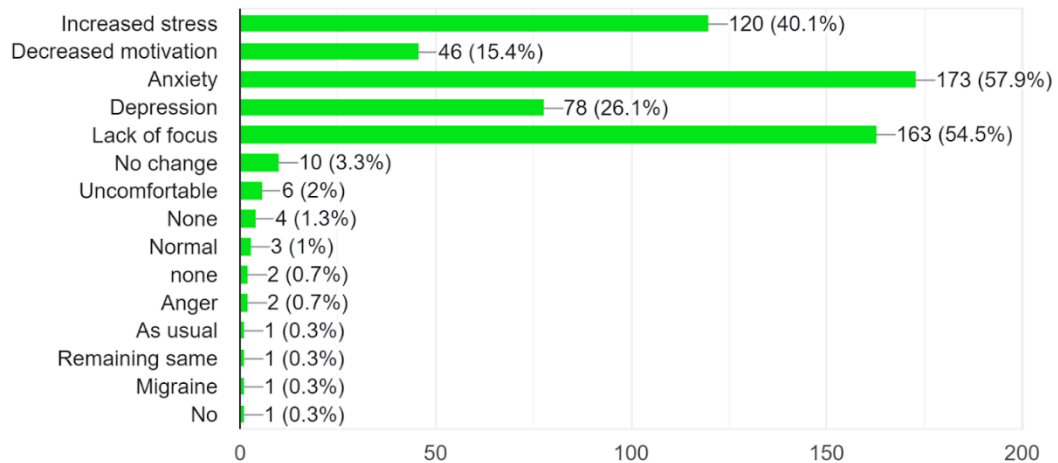


Figure 3.11 Mental health issues of respondents

Comment: A substantial portion of respondents, 57.9%, reported experiencing anxiety, while 54.5% faced a lack of focus. Additionally, 40.1% reported increased stress, 26.1% encountered depression, and 15.4% noted decreased motivation. Interestingly, 3.3% of respondents reported no noticeable changes in their mental health during heatwaves

3.2.12 Impact on sleeping patterns during heatwaves:

Table-3.10: Frequency distribution on impact of sleeping patterns during heatwaves

Impact on sleeping patterns			
	Frequency	Percent	Cumulative percent
Yes	202	67.34	67
No	98	32.67	100
Total	300	100	

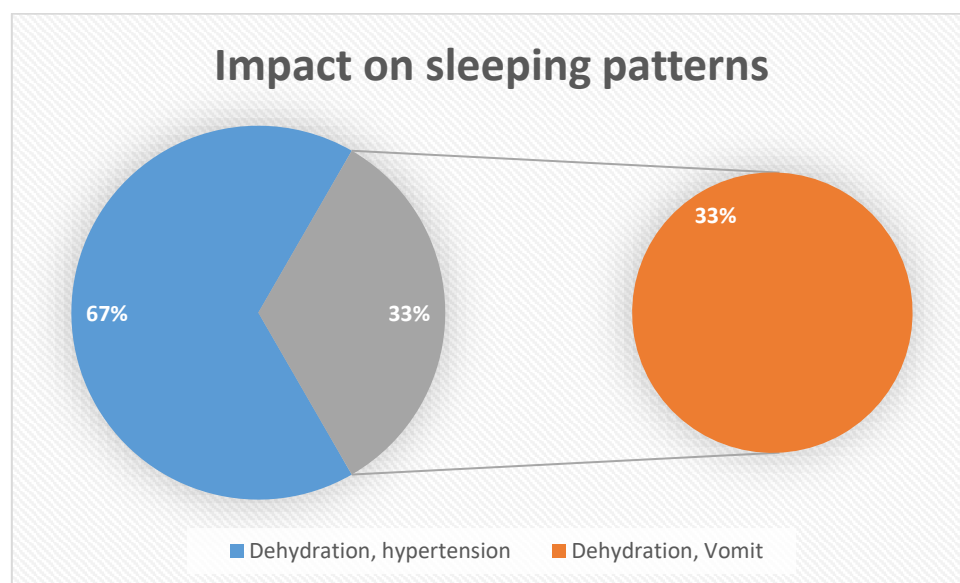


Figure 3.12 Impact on sleeping patterns

Comment: The impact on sleeping patterns during heatwaves is substantial, with 67.34% of respondents indicating disruptions in their sleep. However, it's noteworthy that 32.67% reported no significant impact on their sleep patterns

3.2.13 Changes in eating habits during heatwaves:

Table-3.11: Frequency distribution of eating habits during heatwaves

Changes in eating habits			
	Frequency	Percent	Cumulative percent
Yes, I eat less	150	50	50
Yes, I eat more	21	7	57
No, my eating habits remain the same	129	43	100
Total	300	100	

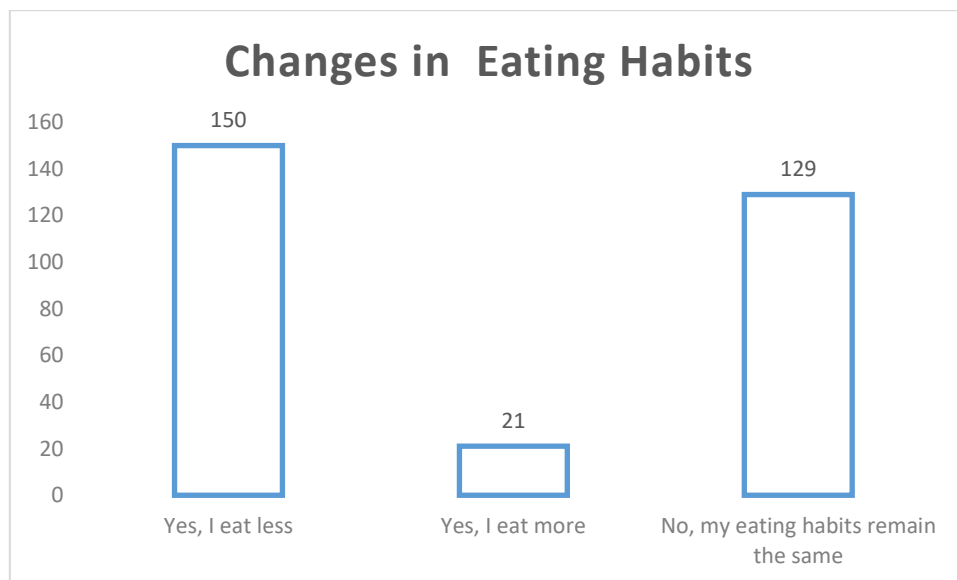


Figure 3.13 Changes in eating habits

Comment: In response to heatwaves, 50% of respondents reported eating less, 7% reported eating more, and 43% mentioned that their eating habits remained the same. This indicates a significant shift in dietary patterns for a considerable portion of the surveyed individuals during heatwave conditions.

3.2.14 Types of foods consume more frequently:

Types of food consume more frequently:

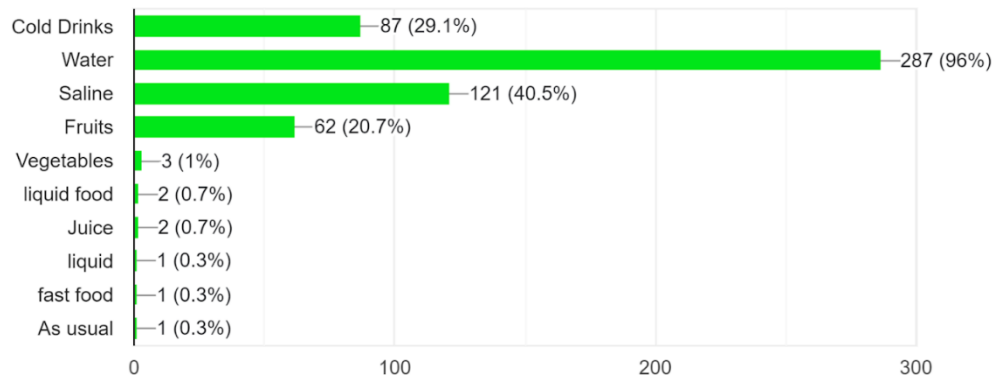


Figure 3.14 Types of foods consume more frequently

Comment: The majority of respondents (96%) reported an increased frequency of water consumption during heatwaves. Additionally, 40.5% of individuals reported consuming saline, 29.1% opted for cold drinks, and 20.7% included fruits in their diet during heatwave conditions. This highlights a diverse range of coping mechanisms related to fluid intake among the surveyed population

3.2.15 How much water consumption during heatwaves:

Table-3.12: Frequency distribution of water consumption during heatwaves

Water Consumption			
	Frequency	Percent	Cumulative Percent
Less than 2 liter	15	5	5
3 liter	68	22.67	27.7
4 liter	126	42	69.7
5 liter	55	18.34	87.8
More than 5 liter	36	12	100
Total	300	100	

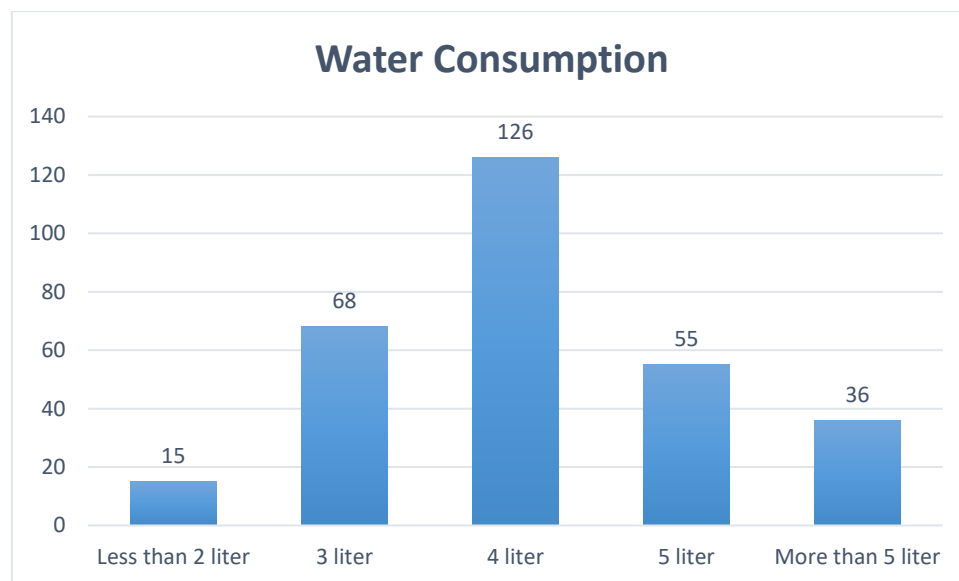


Figure 3.15 water consumption during heatwaves

Comment: The majority of respondents (69.7%) reported consuming 4 liters a day. A significant proportion, 22.67%, opted for 3 liters, while 18.34% reported consuming 5 liters. Interestingly, only 5% reported consuming less than 2 liters, and 12% reported consuming more than 5 liters daily.

3.2.16 Prioritize Dehydration and consume more fluids during heatwaves

Table-3.13: Frequency distribution of prioritize dehydration and consume more fluids

Prioritize Dehydration and consume more fluids				
		Frequency	Percent	Cumulative percent
Valid	Yes, I drink more water	272	90.67	90.5
	No, my fluid intake remains the same	28	9.34	100
	Total	300	100	100

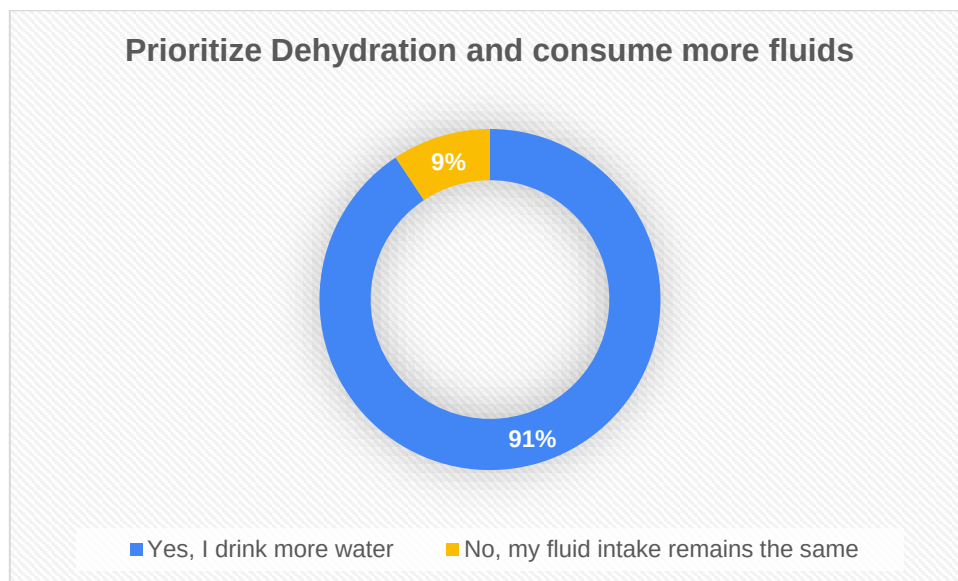


Figure 3.16 Prioritize Dehydration and consume more fluids

Comment: From the above figure we found that 91% respondents drink more water during heat waves and 9% don't care about dehydration.

3.2.17 Performance of work productivity during heatwave:

Table-3.14: Frequency distribution in performance of work productivity

Performance of work productivity			
	Frequency	Percent	Cumulative percent
Yes, my productivity decreases	174	58	58
No, my productivity remains the same	126	42	100
Total	300	100	

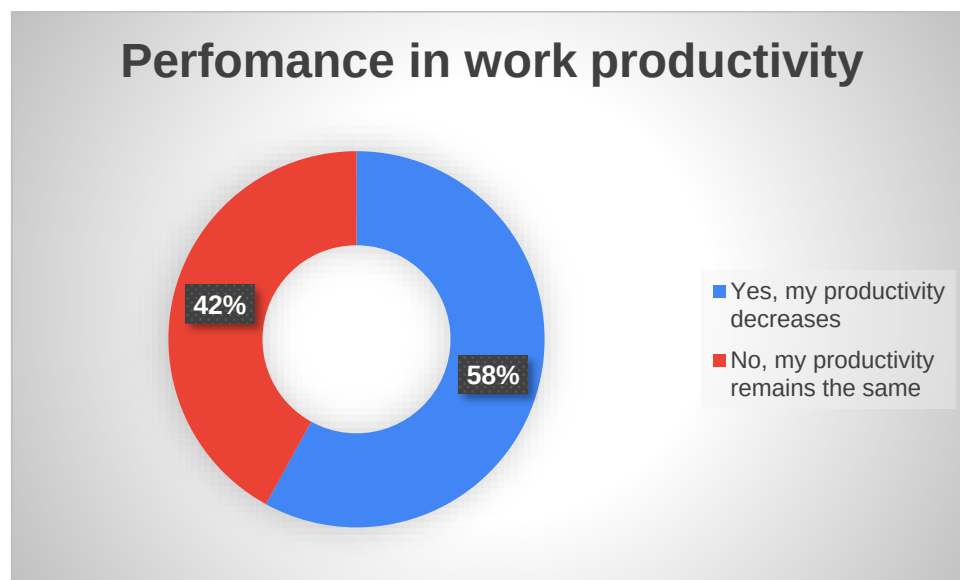


Figure 3.17 Performance in work productivity

Comment: Analyzing work productivity during heatwaves indicates that 58% of respondents experienced a decrease in productivity, while 42% reported no change. These insights underscore the significant impact of heatwaves on individuals' ability to maintain their usual work performance.

3.2.18 Receive any support or accommodations from their employer to cope of heatwaves and maintain work performance

Table-3.15: Frequency distribution of support or accommodations from their employer to cope of heatwaves

support or accommodations from your employer to cope with heatwaves			
	Frequency	Percent	Valid percent
Yes, frequent breaks and access to shaded areas	175	58.33	58
Yes, provision of personal protective equipment (PPE) suitable for heat	8	2.67	61
No, no specific support or accommodations provided	117	39	100
Total	300	100	

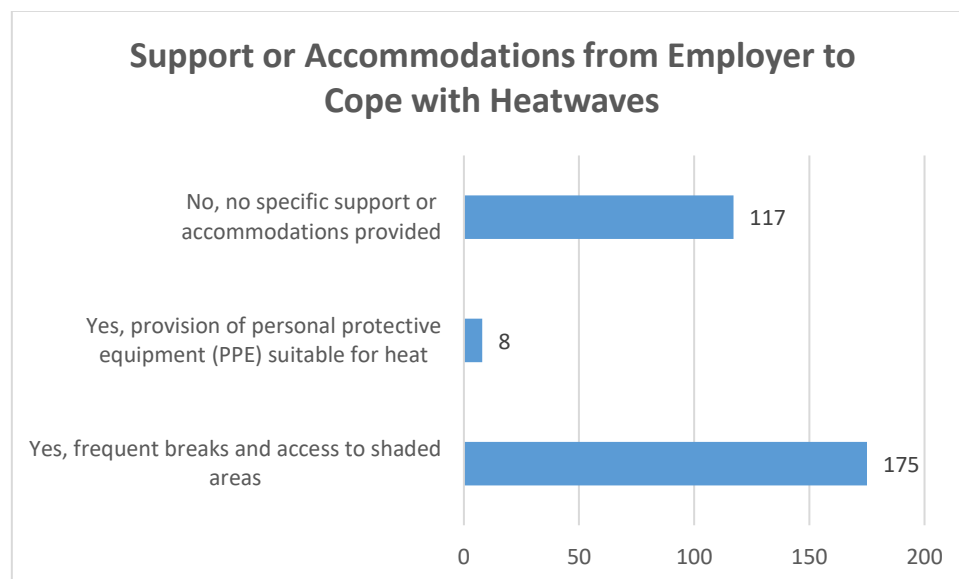


Figure 3.18 support or accommodations from their employer to cope of heatwaves

Comment: Regarding support from employers during heatwaves, 58.33% of respondents reported receiving support such as frequent breaks and access to shaded areas, while 2.67% had provision of personal protective equipment (PPE) suitable for heat. Notably, 39% of participants stated that no specific support or accommodations were provided by their employers during heatwaves.

3.2.19 Effect of heatwaves on monthly income:

Table-3.16: Frequency distribution of effect of heatwaves on monthly income

Effect of heatwaves on monthly income			
	Frequency	Percent	Valid percent
I earn more	21	7	7
I earn less	115	38.34	38
My monthly income remaining same	164	54.67	55
Total	300	100	

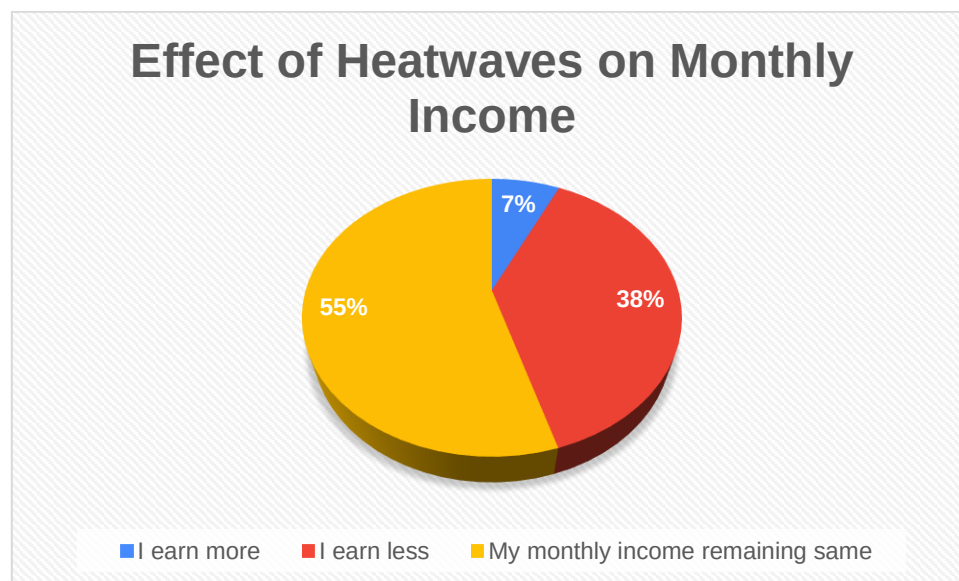


Figure 3.19 Effect of heatwaves on monthly income

Comment: Examining the impact of heatwaves on monthly income, 38.34% of respondents reported earning less during heatwaves, while 54.67% indicated that their monthly income remained the same. Additionally, 7% reported an increase in income during heatwaves.

3.2.20 Financial challenges during heatwaves:

Table-3.17: Frequency distribution of Financial challenges during heatwaves

Financial challenges during heatwaves			
	Frequency	Percent	Valid percent
Yes, increased utility bills (e.g., air-conditioning,fan)	263	87.67	7
Yes, additional costs for personal protective equipment suitable for heat	13	4.34	38
No, no significant financial challenges	24	8	55
Total	300	100	

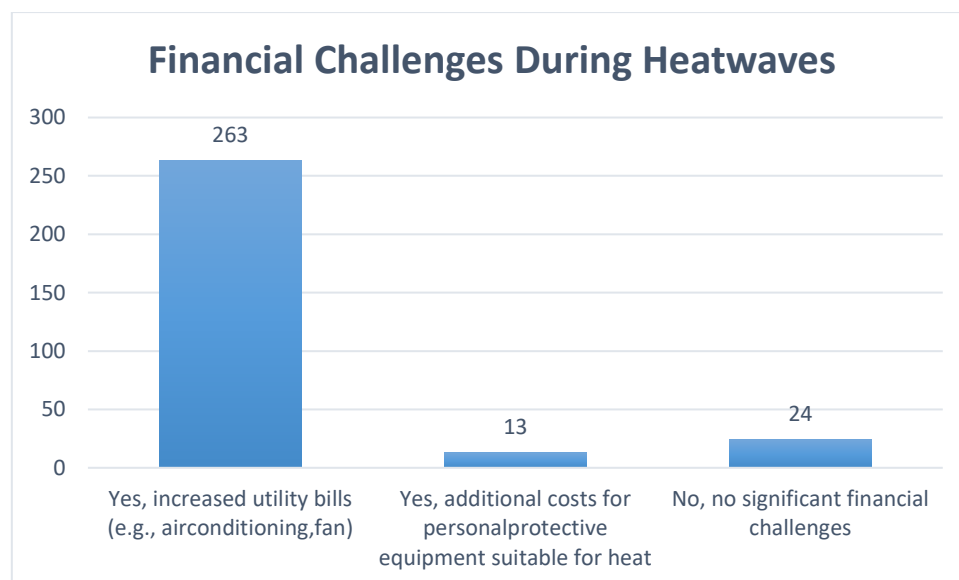


Figure 3.20 Effect of heatwaves on monthly income

Comment: 87.67% experiencing increased utility bills due to items like air-conditioning and fans. A smaller percentage, 4.34%, faced additional costs for personal protective equipment suitable for heat. However, 8% reported no significant financial challenges during heatwaves

3.2.21 Financially prepared to handle unexpected expenses related to heatwaves:

Table-3.18: Frequency distribution of financially prepared to handle unexpected expenses related to heatwaves

Financially prepared to handle unexpected expenses related to heatwaves			
	Frequency	Percent	Cumulative percent
Yes, I have savings or a financial safety net	59	19.67	20
No, I would struggle to handle unexpected expenses	241	80.34	100
Total	300	100	

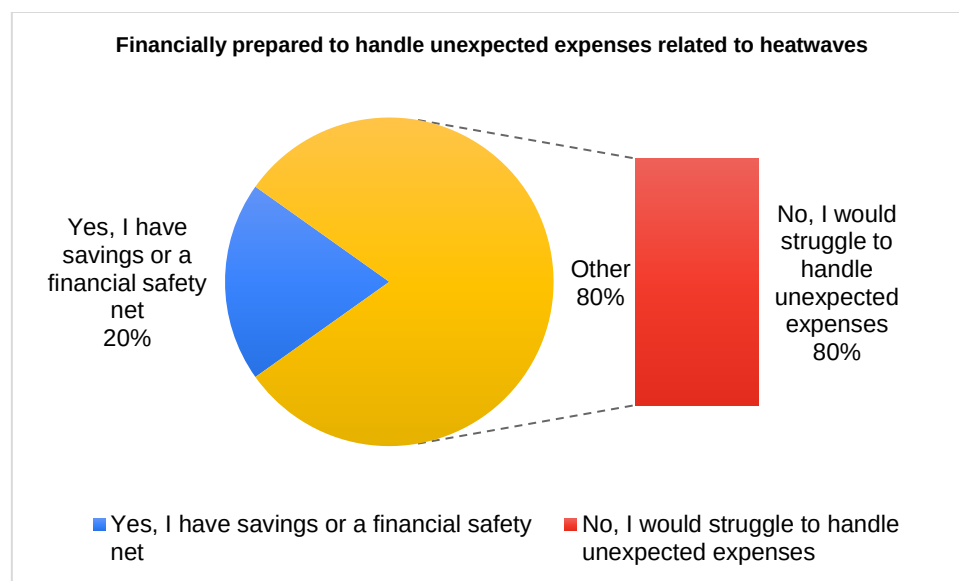


Figure 3.21 Effect of heatwaves on monthly income

Comment: From the above figure we found that 80% have no savings for unexpected expenses i.e they would struggle to handle unexpected expenses and only 20% have a savings for this unwanted circumstances.

3.2.22 Colling facilities

Table-3.19: Frequency distribution of Colling facilities of respondent

Colling facilities			
	Frequency	Percent	Cumulative percent
Yes	291	97	97
No	9	3	100
Total	300	100	

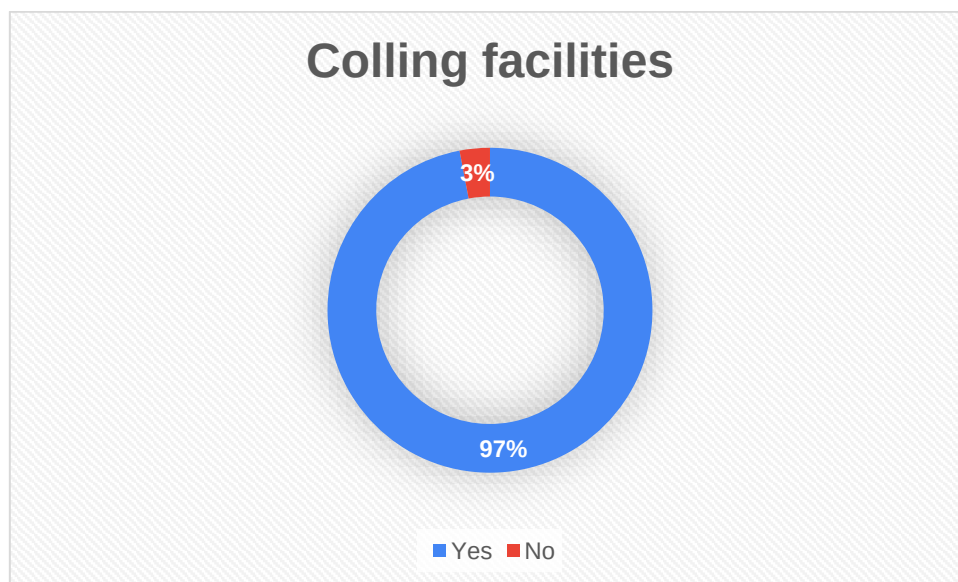


Figure 3.22 Colling facilities of respondent

Comment: An overwhelming 97% of respondents have access to cooling facilities, which include various means such as electric fans, handmade fans, and air conditioning. Only a minor 3% reported having no cooling facilities during heatwaves.

3.2.23 which cooling facilities have used:

Table-3.20: Frequency distribution of Categories of Colling facilities

Cooling facilities			
	Frequency	Percent	Cumulative Percent
Electric fan	51	17	17
Electric fan, Home made fan	128	42.67	59.67
Home made fan	11	3.67	63.34
Electric fan, Table fan	7	2.33	65.67
Home made fan, Table fan	5	1.67	67.34
Electric fan, Air condition	5	1.67	69.01
Charging fan, Air condition	2	0.67	69.68
Electric fan and Charging fan	22	7.33	77.01
Electric fan ,Charging fan and Air condition	10	3.33	80.34
Electric fan, Charging fan, Home made fan	54	18	98.43
Electric fan, Charging fan, Air condition, Home made fan	5	1.67	100
Total	300	100	

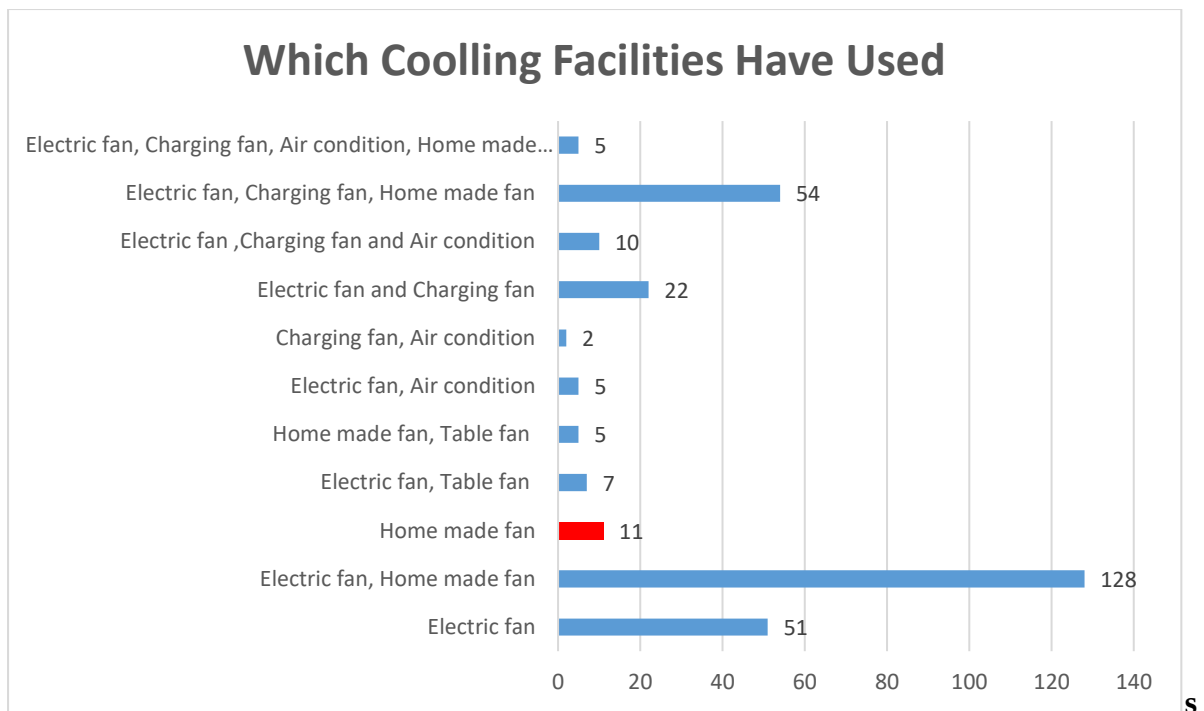


Figure 3.22 Cooling facilities

Comment: Electric fans emerge as the most popular choice, with 17% relying solely on them, while 42.67% opt for a combination of electric and homemade fans. Additionally, 18% utilize a blend of electric fans, homemade fans, and charging fans, demonstrating diverse strategies to combat heat stress among respondents.

Associations between various Quantitative Variables

3.3.1 Pearson Correlation table for finding correlation between Quantitative Variable

Table-3.21: Pearson Correlation table

	BMI	Age	Height	Weight	Working Time
BMI	1	-.026	-.200	.830	.003
Age		1	.031	.001	.197
Height			1	.367	.036
Weight				1	.008
Working Time					1

Interpretation: From this table we can see that there's a **strong positive** correlation between BMI and Weight, suggesting heavier individuals tend to have higher BMIs. Height and Weight show a **moderate positive** correlation, indicating taller individuals may weigh more. Age and Working Time exhibit a weak **positive correlation**, suggesting older individuals may spend more time working. Other correlations are generally weak or negligible, emphasizing nuanced relationships among the variables.

Associations between various Qualitative Variables

1) Associations between **Working Time** and **Effect on mothly income**:

Hypothesis:

H_0 =There is no significant association between **Working Time** and **Effect on mothly income** during heatwaves.

H_1 =There is a significant association between **Working Time** and **Effect on mothly income** during heatwaves.

Table-3.21: Table for Chi-Square Tests

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	42.169 ^a	28	.042
Likelihood Ratio	44.831	28	.023
N of Valid Cases	300		

Comment:

The chi-square test results suggest a statistically **significant association** between **Working Time** and the **Effect on Monthly Income** during heatwaves. With a p-value of 0.042 (less than the typical significance level of 0.05), we reject the null hypothesis. This implies that the number of hours worked (Working Time) is related to the impact on monthly income during heatwaves. In practical terms, it indicates that variations in working hours may contribute to changes in individuals' monthly income when exposed to heatwaves. Further analysis or exploration may be warranted to understand the nature and extent of this association.

2) Associations between **Eating Habit** and **Effect on mothly income**:

Hypothesis:

H_0 =There is no significant association between **Eating Habit** and **Effect on mothly income** during heatwaves.

H_1 =There is a significant association between **Eating Habit** and **Effect on mothly income** during heatwaves.

Table-3.22: Table for Chi-Square Tests

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	12.854	4	.012
Likelihood Ratio	10.141	4	.038
N of Valid Cases	300		

Comment:

The chi-square test results indicate a statistically **significant association** between Eating Habit and the Effect on Monthly Income during heatwaves. The p-value is 0.012, which is below the commonly used significance level of 0.05. Therefore, we reject the null hypothesis, suggesting that there is indeed a significant association between individuals' eating habits and the impact on their monthly income during heatwaves. This implies that changes in eating habits may be linked to variations in monthly income during periods of heatwaves.

3) Association between access to **cooling facilities** and **experiencing physical health issues**:

Hypothesis:

H_0 =There is no significant association between access to cooling facilities and experiencing physical health issues during heatwaves among the surveyed individuals.

H_1 =There is a significant association between access to cooling facilities and experiencing physical health issues during heatwaves among the surveyed individuals.

Table-3.23: Table for Chi-Square Tests

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	.431 ^a	2	.806
Likelihood Ratio	.831	2	.660
N of Valid Cases	300		

Comment:

The chi-square test results suggest that there is no statistically significant association between access to cooling facilities and experiencing physical health issues during heatwaves among the surveyed individuals. Both the Pearson Chi-Square and Likelihood Ratio tests show p-values well above the common significance level of 0.05 (specifically 0.806 and 0.660, respectively). Therefore, we fail to reject the null hypothesis. This implies that, based on the collected data, there isn't sufficient evidence to claim a significant relationship between having access to cooling facilities and the likelihood of experiencing physical health issues during heatwaves among the surveyed individuals.

Model Fitting:

Predicted Model Fitting Analysis:

For fitting a model of our data, we have used **binomial regression analysis**. Which is, In statistics, a binomial logistic regression (often referred to simply as logistic regression), predicts the probability that an observation falls into one of two categories of a dichotomous dependent variable based on one or more independent variables that can be either continuous or categorical. Here the dependent variable should be measured on a dichotomous scale and one or more independent variables, which can be either continuous (i.e., an interval or ratio variable) or categorical (i.e., an ordinal or nominal variable).

4.4.1 Logit Model: Binomial Logistic Regression Model Fitting: For fitting logistic regression model, the dependent variable we choose Having Dining Meal which has two categories coded as

- YES
- NO

For dependent variable we used **impact on sleeping pattern** which is categorical. We used **monthly income** as dependent variable.

Result of Model fitting:

The required computation tables for logistic analysis are given below:

H_0 = There is no significant relationship between dependent variable and selected independent variables.

H_1 =: There is a significant relationship.

Omnibus Tests of Model Coefficients				
		Chi-square	df	Sig.
Step 1	Step	5.967	1	.015
	Block	5.967	1	.015
	Model	5.967	1	.015

Comment: From the above table we can say that, the model assessed using chi-square value statistic, where the p-value (.015) is less than 0.05. So, we may reject the null hypothesis. This proves that there is a significant relationship between the dependent and independent variables.

Hosmer and Lemeshow Test

Hosmer and Lemeshow Test			
Step	Chi-square	df	Sig.
1	9.507	7	.218

Hypothesis:

H_0 : The data fit the model perfectly.

H_1 : The data doesn't fit the model perfectly.

Comment: From the above table we can say that, the model assessed using chi-square value statistic, where the p-value (0.218) is greater than 0.05. So, we may accept the null hypothesis. This proves that the data fit the model perfectly.

Chapter Four

Summary and Conclusion:

The survey on the multifaceted impact of heatwaves on human health in Pabna City, Bangladesh, reveals significant findings across demographic, health, lifestyle, and socio-economic dimensions. The study, conducted with a sample of 300 respondents, sheds light on the challenges faced during heatwaves and emphasizes the prevalence of dehydration, intersecting with various health concerns. Mental health emerges as a critical dimension, notably affected by anxiety and decreased focus.

Demographically, the sample is predominantly male with diverse educational backgrounds and occupations, falling mostly within the middle socio-economic category. The survey indicates a high percentage (96.67%) of respondents experiencing health changes during heatwaves, with dehydration, fever, and skin problems being the predominant physical health issues.

The impact on mental health is substantial, affecting 92% of respondents, with anxiety and decreased focus being prevalent issues. Lifestyle changes, including alterations in sleeping patterns and eating habits, are reported by 67% and 50% of respondents, respectively.

Financial repercussions are evident, with 38% experiencing a decrease in monthly income, and 87.67% facing increased utility bills during heatwaves. Employer support during heatwaves varies, with 58.33% receiving frequent breaks and access to shaded areas.

In conclusion, this survey contributes a comprehensive understanding of the impact of heatwaves on human health in Pabna City, encompassing physical, mental, and socio-economic dimensions. The findings underscore the urgent need for targeted interventions and policy considerations to mitigate the adverse effects of heatwaves on the population's health and well-being. The study's specific focus on a smaller urban center in a developing country provides valuable insights into the unique challenges faced by such communities, addressing a research gap in the existing literature. The results can serve as a foundation for informed decision-making and the development of effective strategies to protect vulnerable groups during extreme heat events.

Further Study:

1. **Climatic and Health Impact Modeling:** Develop models that simulate the potential future scenarios of heatwaves in Pabna City based on climate change projections. Integrate these models with health impact assessments to predict and quantify the future burden on public health.
2. **Comparative Analysis with Global Urban Centers:** Undertake a comparative analysis between the impacts of heatwaves in Pabna City and other global urban centers facing similar climate challenges. Identify commonalities and differences to contribute to a broader understanding of the global implications of urban heatwaves.
3. **Psychosocial Impacts of Heatwaves:** Delve deeper into the psychosocial impacts of heatwaves, with a particular focus on anxiety and decreased focus. Investigate the underlying psychological mechanisms, coping strategies employed by the population, and potential interventions to mitigate mental health effects.
4. **Gendered Impacts of Heatwaves:** Explore the gendered impacts of heatwaves, considering differential vulnerabilities and coping mechanisms. Analyze how gender dynamics intersect with socio-economic factors to influence health outcomes during extreme heat events.

Application:

1. **Community Awareness and Education:** Develop a comprehensive awareness campaign to educate residents about the health risks associated with heatwaves and promote adaptive behaviors. This can include workshops, informational materials, and digital communication strategies to reach a wide audience.
2. **Community Engagement and Feedback Mechanism:** Establish a community engagement platform to gather feedback on the effectiveness of implemented measures and to address evolving community needs. Use this mechanism to refine and improve the initiative based on real-time experiences and challenges.

Limitation:

1. **Sample Size and Representativeness:** The survey's sample size of 300 respondents, while providing valuable insights, may not fully represent the diversity of Pabna City's population. It might not capture the experiences of certain demographic groups or individuals with specific vulnerabilities.
2. **Demographic Bias:** The predominantly male composition of the sample (75.34%) could introduce a gender bias. The findings might not fully reflect the unique challenges faced by women or other gender groups during heatwaves.
3. **Single Geographic Focus:** The study focuses solely on Pabna City, limiting its generalizability to other urban or rural areas. Different geographic regions may have unique climate characteristics, socio-economic factors, and health challenges during heatwaves.

References:

1. Nori-Sarma, A., Anderson, G. B., Rajiva, A., ShahAzhar, G., Gupta, P., Pednekar, M. S., ... & Bell, M. L. (2019). The impact of heat waves on mortality in Northwest India. *Environmental research*, 176, 108546.
2. Akhtar, R. (2020). Introduction: extreme weather events and human health: a global perspective. *Extreme Weather Events and Human Health: International Case Studies*, 3-11.
3. Kilbourne, E. M. (1989). Heatwaves. *The public health consequences of disasters*, 51-61.
4. Cochran, W. G. (1942). Sampling theory when the sampling-units are of unequal sizes. *Journal of the American Statistical Association*, 37(218), 199-212.
5. Islam, M.N (2006) Research Methodology, 2nd Edition, Mullick and Brothers, Dhaka.
6. Mood, A. M., Graybill, F. A. Boes, D. C. (1974): Introduction to the theory of statistics, 3rd edition, McGraw-Hill, New York.
7. Gujarati, D. N. (2003). Basic Econometrics, 4th edition, McGraw Hill, New York.
8. Hobday, A. J., Alexander, L. V., Perkins, S. E., Smale, D. A., Straub, S. C., Oliver, E. C., ... & Wernberg, T. (2016). A hierarchical approach to defining marine heatwaves. *Progress in Oceanography*, 141, 227-238.
9. Perkins, S. E. (2015). A review on the scientific understanding of heatwaves—their measurement, driving mechanisms, and changes at the global scale. *Atmospheric Research*, 164, 242-267.

Appendix

SPSS= Statistical Package for the Social Sciences

USA= United States of America

WHO= World Health Organization

Field Survey Questionnaire

Pabna University of Science and Technology



Department of Statistics Field Survey Questionnaire

On

“Impact of heatwaves on human health: A Case Study in Pabna”

Hello, we are working with a survey group, Department of Statistics, PUST. We are conducting a survey about “Impact of Heatwaves on Human Health: A Case Study in Pabna”. We need some information’s about your complications of heatwaves for our filed survey. All of the answers you are given that will be confidential and will not be shared with anyone other than members of our survey team. If you agree, please fill up the questionnaire.

1. Name:

2. Sex: ☐ Male ☐ Female

3. Age (in year) 4. Height (feet) (Inch)..... 5. Weight (in kg).....

6. Level of education:

☐ Illiterate ☐ Primary ☐ Secondary ☐ Higher Secondary ☐ Graduate ☐ Others.....

7. Marital status: ☐ Married ☐ Unmarried ☐ Divorced ☐ Widowed

8. Occupation: ☐ Farmer ☐ Day labour ☐ Construction Worker ☐ Rickshawpuller ☐ Student

☐ Service Holder ☐ Sales man ☐ Others.....

9. Socio economic status: ☐ Poorest ☐ Poor ☐ Middle ☐ Rich ☐ Richest

10. How many Time do you work at your work place: (In hours)

11. Do you have any health changes for heatwaves than from normal weather?

☐ Yes ☐ No

12. Have you experienced any physical health issues during heatwaves? ☐ Yes ☐ No

13. If yes, please select the physical health issues you have experienced during heatwaves:

☐ Dehydration ☐ Heatstroke ☐ Fever ☐ Vomit ☐ skin problems ☐ others (please specify).....

14. Have you noticed any changes in your mental health during heatwaves?

☐ Yes ☐ No

15. If yes, please select the mental health issues you have experienced during heatwaves:

☐ Increased stress ☐ Decreased motivation ☐ Anxiety ☐ Depression ☐ Lack of focus

☐ Other (please specify).....

16. Did the heatwaves impact your sleeping patterns?

☐ Yes ☐ No

17. Do you experience changes in your eating habits during heatwaves?

☐ Yes, I eat less ☐ Yes, I eat more ☐ No, and my eating habits remain the same

18. If you experience changes in your eating habits during heatwaves, please select the types of food you consume more frequently:

☐ Cold Drinks ☐ Water ☐ Saline ☐ Fruits ☐ others (please specify).....

19. How much water you consume during heatwaves?

☐ Less than 2 liter ☐ 3 liter ☐ 4 liter ☐ 5 liter ☐ More than 5 liter

20. Do you prioritize hydration and consume more fluids during heatwaves?

☐ Yes, I drink more water ☐ No, my fluid intake remains the same

21. How does the heatwave affect your work performance?

☐ Decreased energy levels ☐ Difficulty concentrating

☐ Reduced physical stamina ☐ No noticeable impact

22. Have you observed a decrease in your work productivity during heatwaves?

☐ Yes, my productivity decreases ☐ No, my productivity remains the same

23. Do you receive any support or accommodations from your employer to cope with heatwaves and maintain work performance?

☐ Yes, frequent breaks and access to shaded areas

☐ Yes, provision of personal protective equipment (PPE) suitable for heat

☐ No, no specific support or accommodations provided

24. Did heatwaves effect your monthly income?

☐ I earn more ☐ I earn less ☐ remaining same

25. Do you face any financial challenges during heatwaves?

☐ Yes, increased utility bills (e.g., air conditioning, fan)

☐ Yes, additional costs for personal protective equipment suitable for heat

☐ No, no significant financial challenges

26. Are you financially prepared to handle unexpected expenses related to heatwaves?

☐ Yes, I have savings or a financial safety net

☐ No, I would struggle to handle unexpected expenses

27. Did you have access to colling facilities?

☐ Yes ☐ No

28. If yes, which colling facilities you have used?

☐ Electric fan ☐ Charging fan ☐

Air condition ☐ Home made fan

Thank you for completing this
questionnaire.

.....
Signature of the respondent

Name of the
interviewer.....
Date.....

